tp_imp_mult

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Statistique univarié des données

```
summary(d)
```

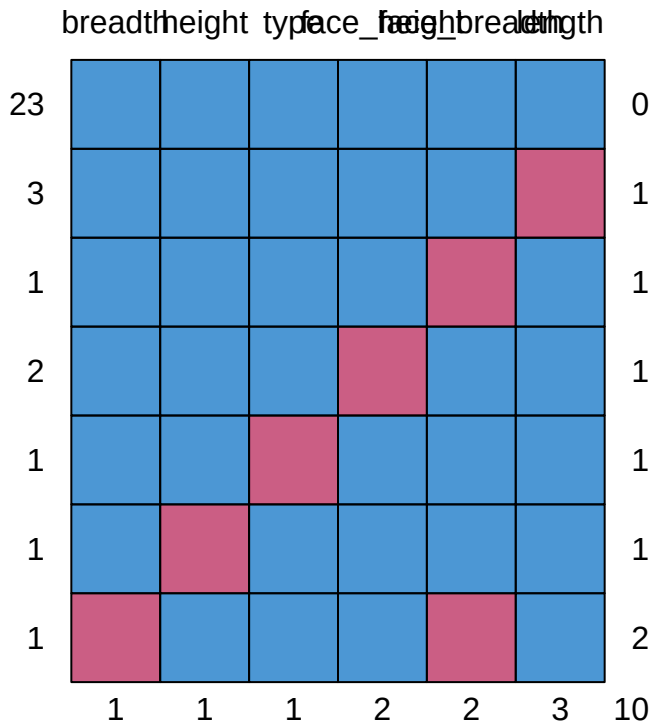
```
##      length         breadth          height       face_height
##  Min.   :162.5   Min.   :126.5   Min.   :121.5   Min.   :62.00
##  1st Qu.:173.5   1st Qu.:134.8   1st Qu.:129.2   1st Qu.:69.62
##  Median :179.5   Median :139.0   Median :133.5   Median :73.50
##  Mean   :180.6   Mean   :139.2   Mean   :133.3   Mean   :72.87
##  3rd Qu.:185.0   3rd Qu.:142.5   3rd Qu.:137.2   3rd Qu.:76.50
##  Max.   :200.0   Max.   :153.0   Max.   :145.0   Max.   :82.50
##  NA's   :3       NA's   :1       NA's   :1       NA's   :2
##   face_breadth     type
##  Min.   :118.5   A   :16
##  1st Qu.:131.6   B   :15
##  Median :134.5   NA's: 1
##  Mean   :133.8
##  3rd Qu.:136.5
##  Max.   :146.5
##  NA's   :2
```

On remarque que toutes less variables ont des valeurs manquantes (3 pour length, 1 pour breadth et height et 2 pour face_height et face_breadth)

Statistiques bivariées

```
Length :
    Le test t de Welch sur la longueur (length) montre une différence significative entre les groupes A

Breadth :
    Le test t de Welch sur la largeur (breadth) ne montre pas de différence significative entre les grou

Height :
    Le test t de Welch sur la hauteur (height) ne montre pas de différence significative entre les group

Face Height :
    Le test t de Welch sur la hauteur du visage (face_height) montre une différence significative entre

Face Breadth :
    Le test t de Welch sur la largeur du visage (face_breadth) montre une différence significative entre
```

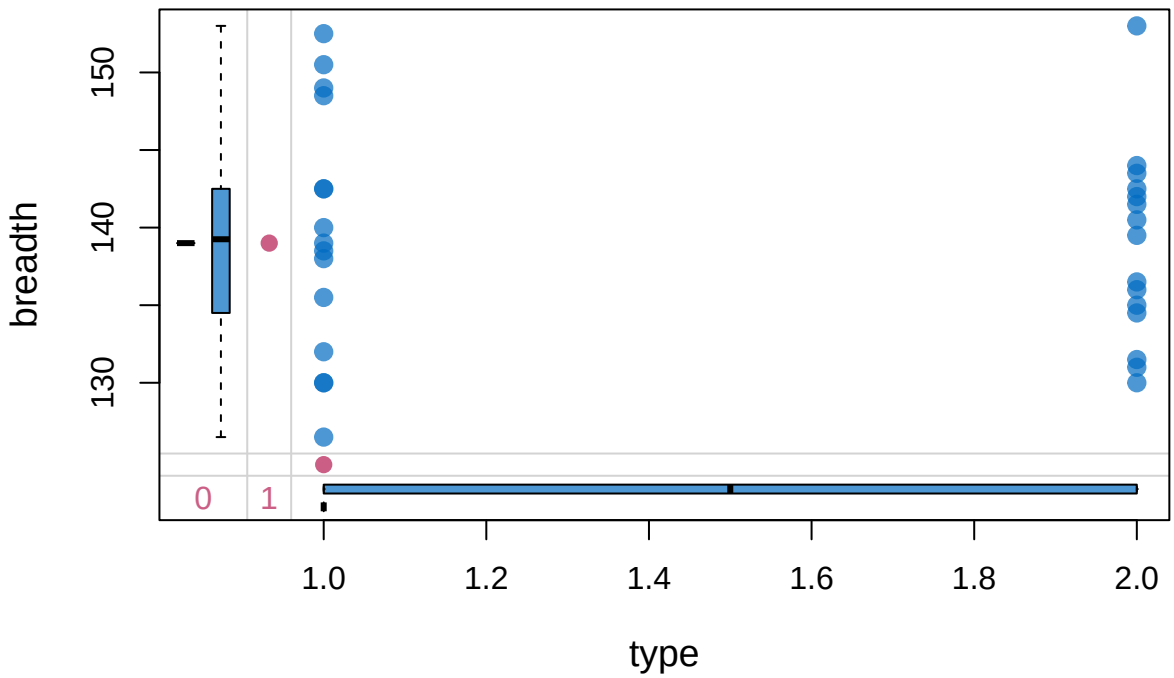
```
md.pattern(d)
```



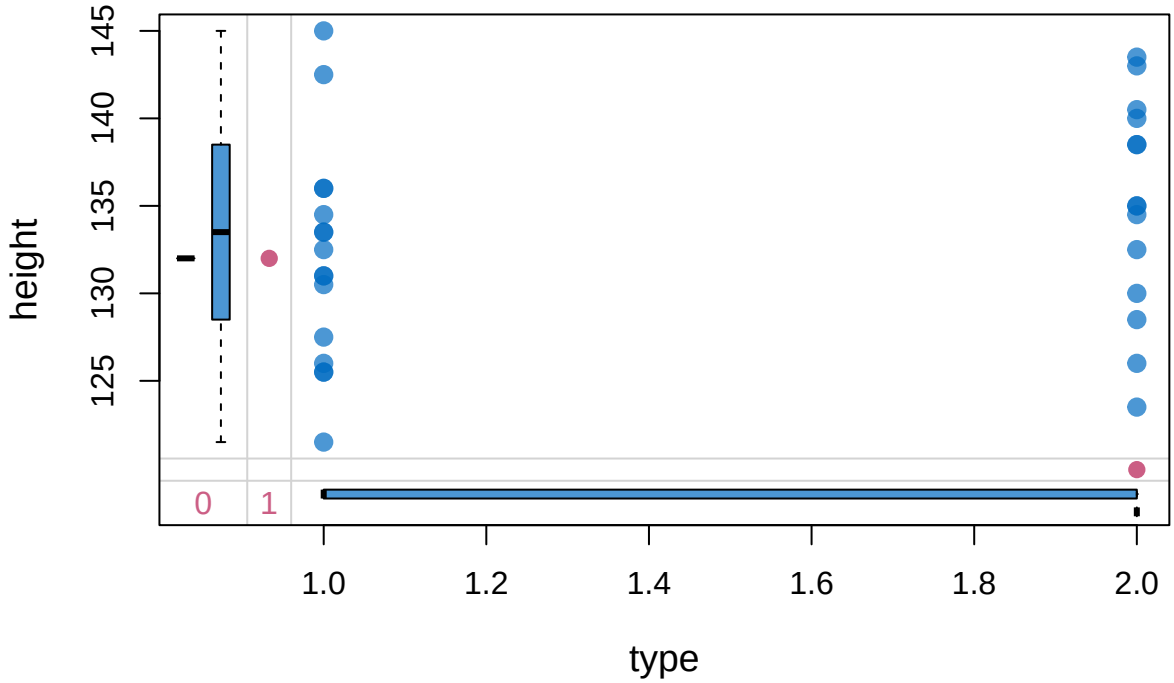

```
##    breadth height type face_height face_breadth length
## 23       1      1    1           1            1      1  0
## 3        1      1    1           1            1      0  1
## 1        1      1    1           1            0      1  1
## 2        1      1    1           0            1      1  1
## 1        1      1    0           1            1      1  1
## 1        1      0    1           1            1      1  1
## 1        0      1    1           1            0      1  2
##          1      1    1           2            2      3 10
```

On observe qu'il y à 23 observations avec 0 donnéess manquantes, 3 avec 1 pour lenghts, 1 avec 1 pour face_breath, 2 avec 1 pourface_height, 1 avec 1 pour type, 1 avec 1 pour height et 1 avec 2 pour breadth et face_breadth

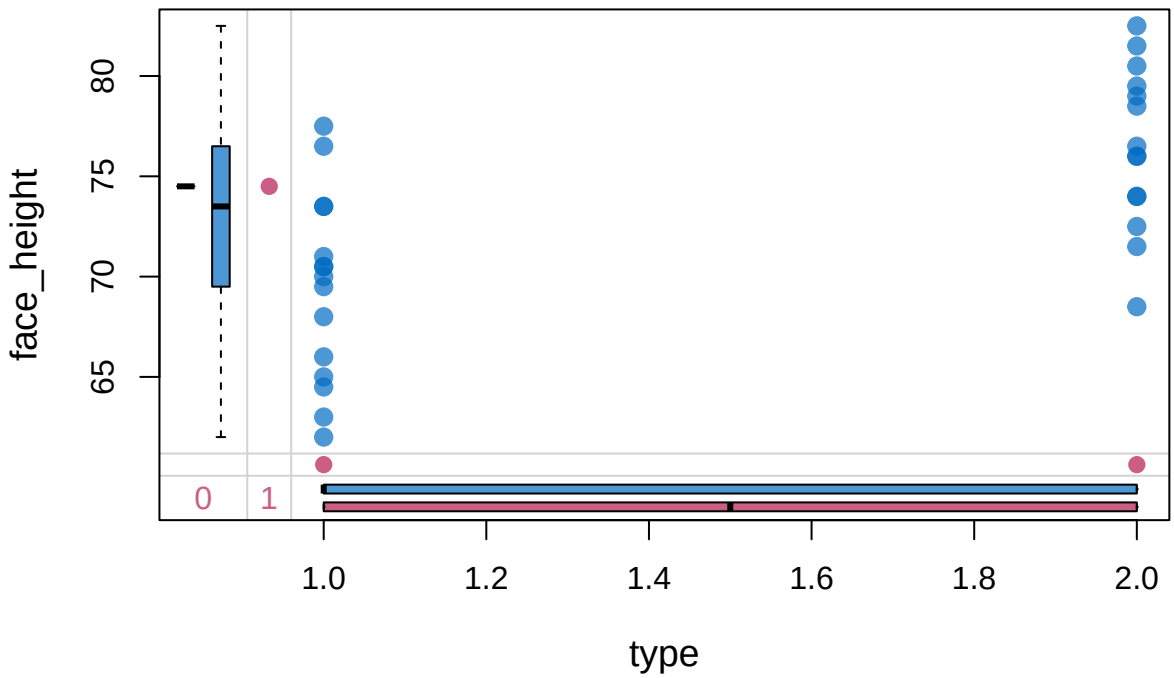
```
marginplot(tib_skul[, c("type","breadth")], col = mdc(c("obs", "mis")), cex = 1.2, cex.lab = 1.2,pch=19
```



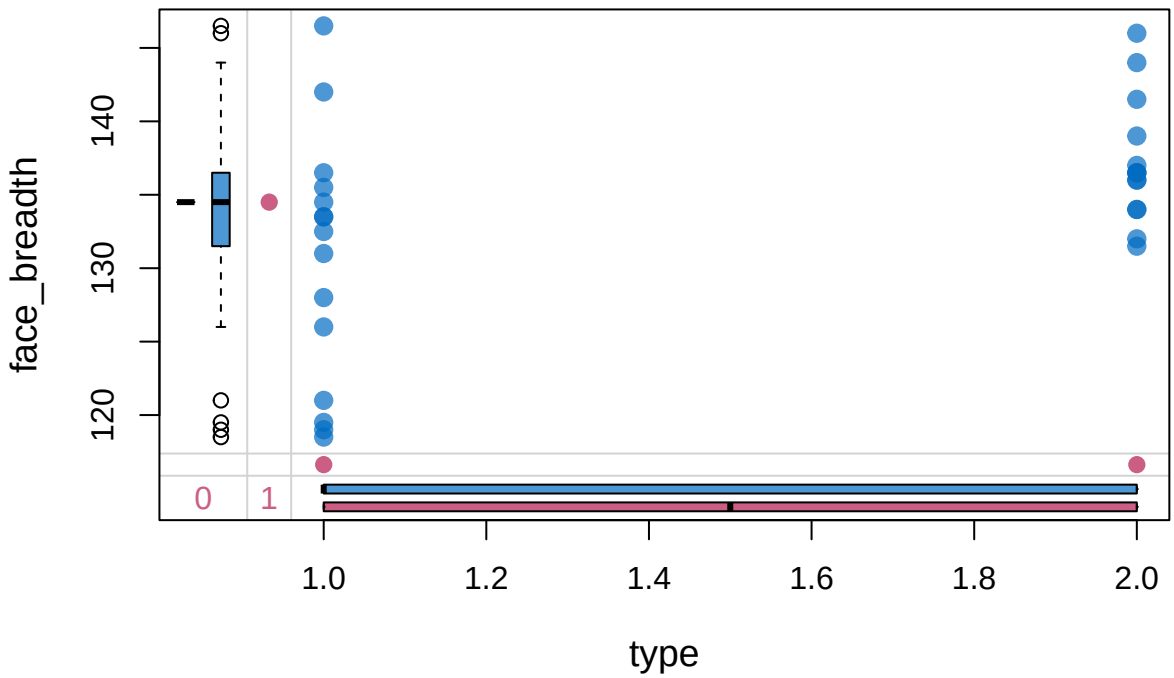

```
marginplot(tib_skul[, c("type","height")], col = mdc(c("obs", "mis")), cex = 1.2, cex.lab = 1.2,pch=19)
```



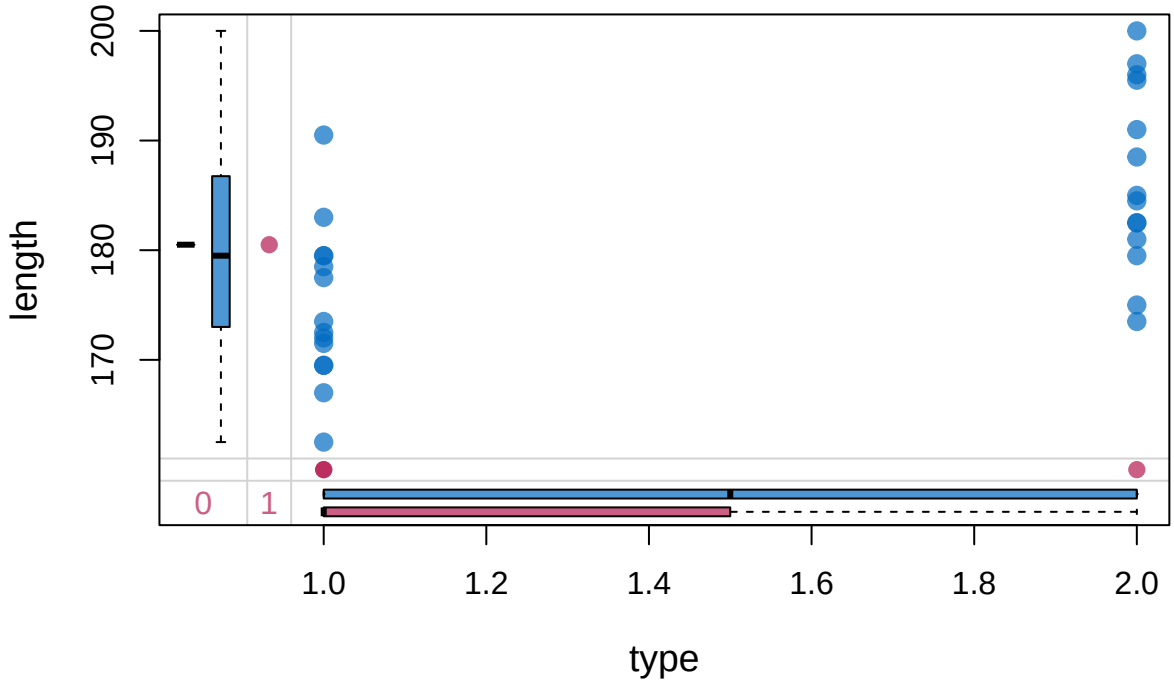

```
marginplot(tib_skul[, c("type","face_height")], col = mdc(c("obs", "mis")), cex = 1.2, cex.lab = 1.2,pch
```




```
marginplot(tib_skul[, c("type","face_breadth")], col = mdc(c("obs", "mis")), cex = 1.2, cex.lab = 1.2,p
```




```
marginplot(tib_skul[, c("type","length")], col = mdc(c("obs", "mis")), cex = 1.2, cex.lab = 1.2,pch=19)
```



Boxplot assez proche (pour une valeur manquante) on peut utiliser MICE

Imputation des données manquantes

Regression logistique

On complete Donnéess manquantes avec MICE:

```
dm = mice(d, m = 5, seed=10, print = FALSE, maxit = 50)
summary(dm)
```

```
## Class: mids
## Number of multiple imputations:  5
## Imputation methods:
##       length      breadth       height  face_height face_breadth         type
##        "pmm"        "pmm"        "pmm"        "pmm"        "pmm"     "logreg"
## PredictorMatrix:
##              length breadth height face_height face_breadth type
## length            0       1      1           1             1    1
## breadth           1       0      1           1             1    1
## height            1       1      0           1             1    1
## face_height       1       1      1           0             1    1
## face_breadth      1       1      1           1             0    1
## type              1       1      1           1             1    0
```

La méthode pmm est utilisé pour imputer les différentes variables en se basant sur la moyenne prédite

On peut observer les données imputées:

```r
print(dm$imp)
```

```
## $length
##        1     2     3     4     5
## 5  196.0 173.5 191.0 169.5 180.5
## 17 162.5 173.5 172.5 173.5 162.5
## 29 180.5 175.0 191.0 182.5 183.0
##
## $breadth
##        1   2     3     4     5
## 11 131.5 142 142.5 141.5 142.5
##
## $height
##        1     2     3   4     5
## 22 127.5 133.5 140.5 135 140.5
##
## $face_height
##       1    2    3    4  5
## 6  70.5 77.5 74.5 73.5 65
## 21 76.0 81.5 68.5 79.0 76
##
## $face_breadth
##        1     2     3   4   5
## 11 119.5 146.5 135.5 131 128
## 25 136.5 141.5 144.0 134 146
##
## $type
##    1 2 3 4 5
## 13 B B B B B
```

```r
d1 = complete(dm,1)
summary(d1)
```

```
##      length         breadth          height       face_height
##  Min.   :162.5   Min.   :126.5   Min.   :121.5   Min.   :62.00
##  1st Qu.:173.2   1st Qu.:133.9   1st Qu.:128.2   1st Qu.:69.88
##  Median :180.0   Median :139.0   Median :133.0   Median :73.50
##  Mean   :180.5   Mean   :139.0   Mean   :133.1   Mean   :72.89
##  3rd Qu.:185.9   3rd Qu.:142.5   3rd Qu.:136.6   3rd Qu.:76.50
##  Max.   :200.0   Max.   :153.0   Max.   :145.0   Max.   :82.50
##   face_breadth   type
##  Min.   :118.5   A:16
##  1st Qu.:131.4   B:16
##  Median :134.5
##  Mean   :133.4
##  3rd Qu.:136.5
##  Max.   :146.5
```

```r
d2 = complete(dm,2)
summary(d2)
```

```
##      length         breadth          height       face_height
##  Min.   :162.5   Min.   :126.5   Min.   :121.5   Min.   :62.00
##  1st Qu.:173.5   1st Qu.:134.9   1st Qu.:129.6   1st Qu.:69.88
##  Median :179.5   Median :139.2   Median :133.5   Median :73.75
```

```
##  Mean   :180.0   Mean   :139.3   Mean   :133.3   Mean   :73.28
##  3rd Qu.:184.6   3rd Qu.:142.5   3rd Qu.:136.6   3rd Qu.:77.50
##  Max.   :200.0   Max.   :153.0   Max.   :145.0   Max.   :82.50
##   face_breadth   type
##  Min.   :118.5   A:16
##  1st Qu.:131.9   B:16
##  Median :135.0
##  Mean   :134.4
##  3rd Qu.:137.5
##  Max.   :146.5
```

```
d3 = complete(dm,3)
summary(d3)
```

```
##      length         breadth          height       face_height
##  Min.   :162.5   Min.   :126.5   Min.   :121.5   Min.   :62.00
##  1st Qu.:173.2   1st Qu.:134.9   1st Qu.:129.6   1st Qu.:69.25
##  Median :180.0   Median :139.2   Median :133.5   Median :73.50
##  Mean   :181.0   Mean   :139.3   Mean   :133.5   Mean   :72.78
##  3rd Qu.:189.0   3rd Qu.:142.5   3rd Qu.:138.5   3rd Qu.:76.50
##  Max.   :200.0   Max.   :153.0   Max.   :145.0   Max.   :82.50
##   face_breadth   type
##  Min.   :118.5   A:16
##  1st Qu.:131.9   B:16
##  Median :135.0
##  Mean   :134.1
##  3rd Qu.:136.6
##  Max.   :146.5
```

```
d4 = complete(dm,4)
summary(d4)
```

```
##      length         breadth          height       face_height
##  Min.   :162.5   Min.   :126.5   Min.   :121.5   Min.   :62.00
##  1st Qu.:173.2   1st Qu.:134.9   1st Qu.:129.6   1st Qu.:69.88
##  Median :179.5   Median :139.2   Median :133.5   Median :73.50
##  Mean   :180.1   Mean   :139.3   Mean   :133.4   Mean   :73.08
##  3rd Qu.:184.6   3rd Qu.:142.5   3rd Qu.:136.6   3rd Qu.:76.75
##  Max.   :200.0   Max.   :153.0   Max.   :145.0   Max.   :82.50
##   face_breadth   type
##  Min.   :118.5   A:16
##  1st Qu.:131.4   B:16
##  Median :134.2
##  Mean   :133.7
##  3rd Qu.:136.5
##  Max.   :146.5
```

```
d5 = complete(dm,5)
summary(d5)
```

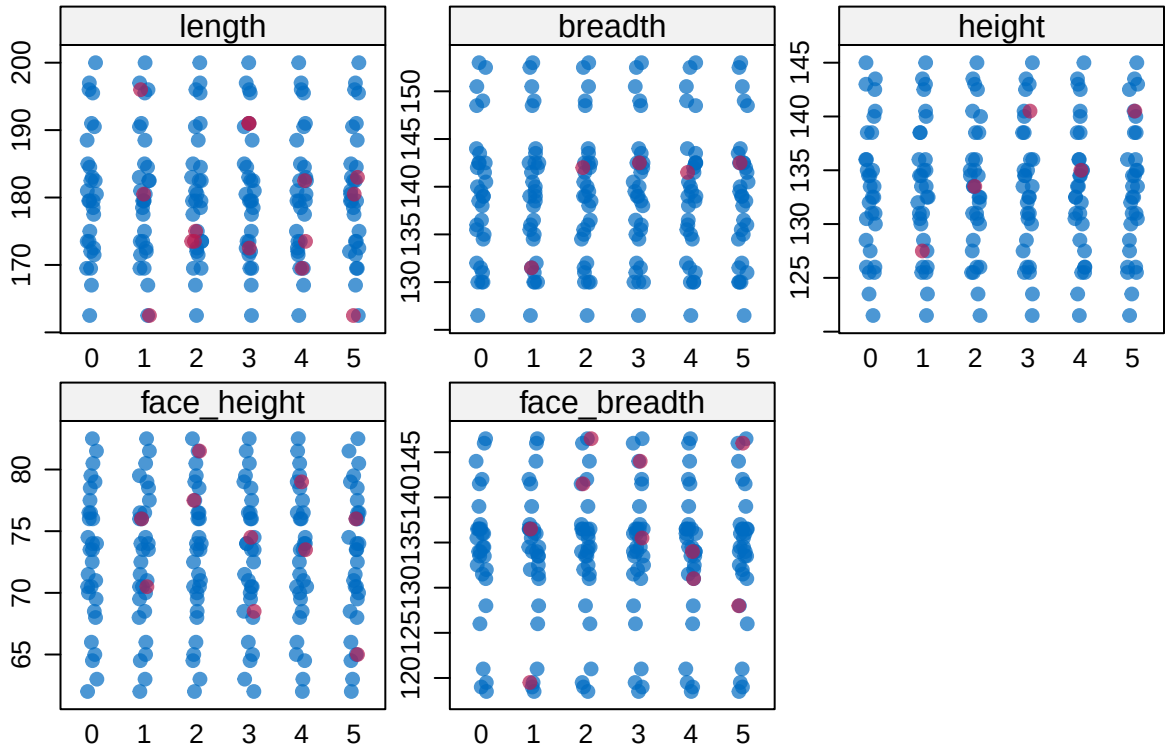
```
##      length         breadth          height       face_height
##  Min.   :162.5   Min.   :126.5   Min.   :121.5   Min.   :62.00
##  1st Qu.:173.2   1st Qu.:134.9   1st Qu.:129.6   1st Qu.:69.25
##  Median :180.0   Median :139.2   Median :133.5   Median :73.50
##  Mean   :180.1   Mean   :139.3   Mean   :133.5   Mean   :72.72
##  3rd Qu.:184.6   3rd Qu.:142.5   3rd Qu.:138.5   3rd Qu.:76.50
```

```
##  Max.   :200.0   Max.   :153.0   Max.   :145.0   Max.   :82.50
##   face_breadth   type
##  Min.   :118.5   A:16
##  1st Qu.:131.4   B:16
##  Median :134.5
##  Mean   :134.0
##  3rd Qu.:136.6
##  Max.   :146.5
```

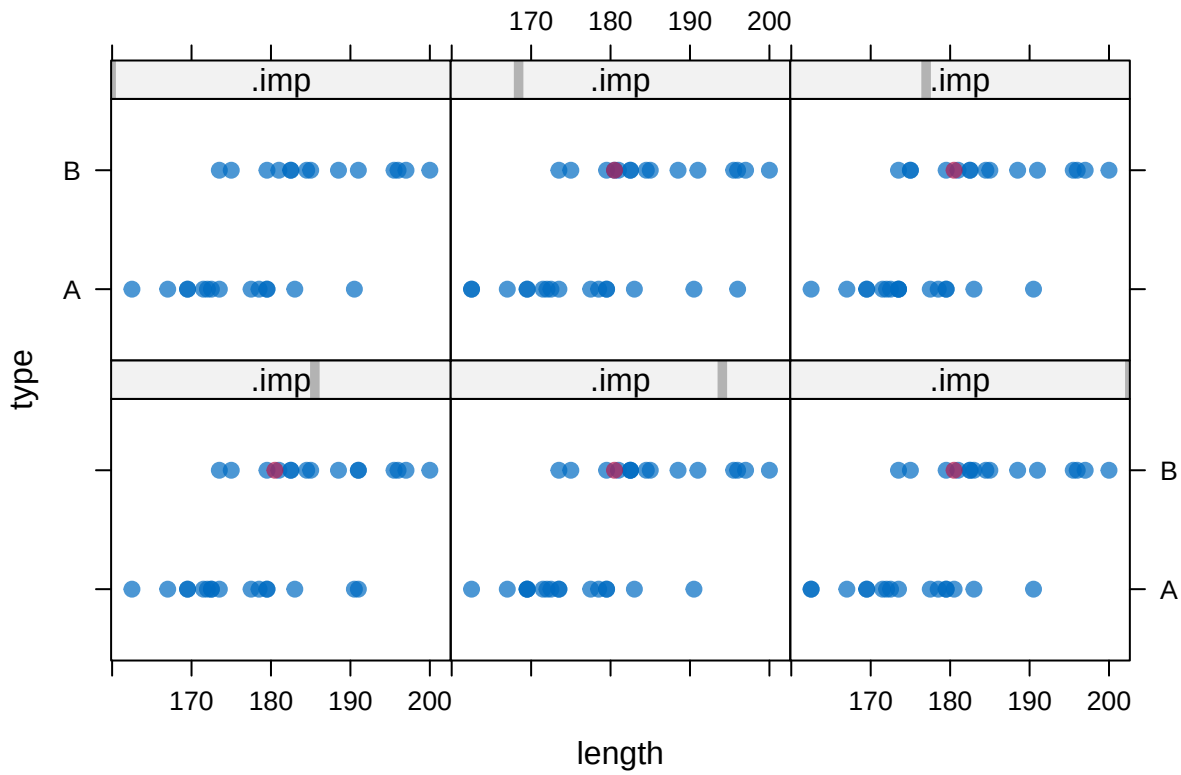
On observe une très légère différence au niveau de la moyenne et de la médiane pour les 5 différents jeux

Représentation graphique des données imputées:

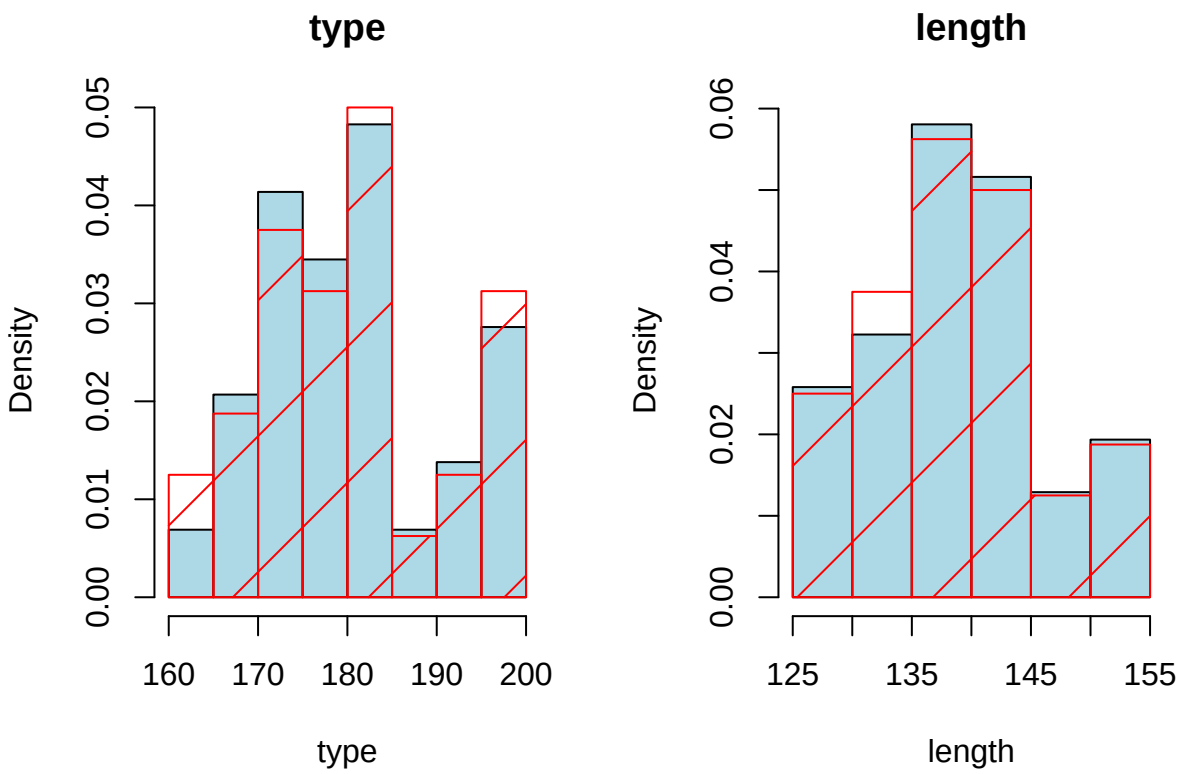
```
stripplot(dm, pch = 20, cex = 1.2)
```




```
xyplot(dm, type ~ length|.imp, pch = 20, cex = 1.4)
```

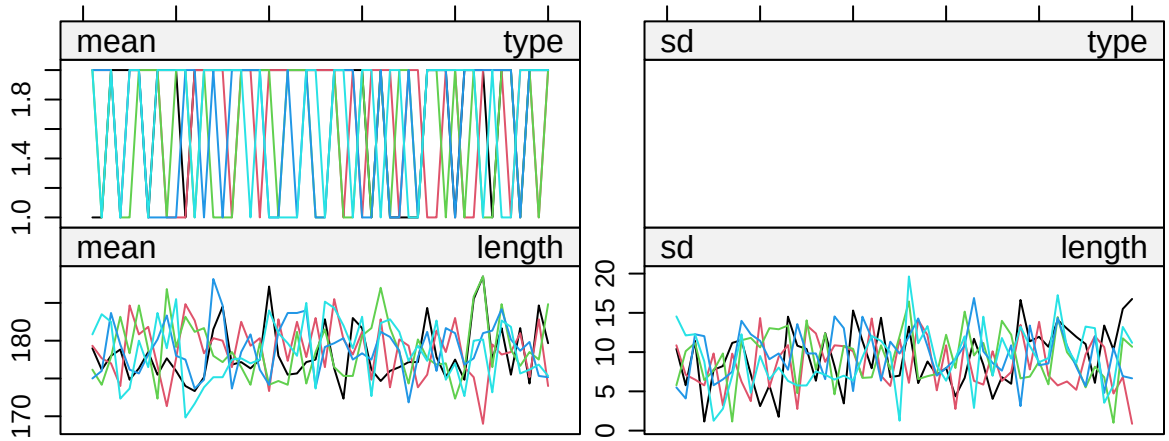



```
par(mfrow=c(1,2))
hist(d[,1], col="lightblue", probability = TRUE, xlab = "type", main = "type")
hist(d1[,1], density = 2, add= TRUE, border = "red", probability = TRUE, col ="red")

hist(d[,2], col="lightblue", probability = TRUE, xlab = "length", main = "length")
hist(d1[,2], density = 2, add= TRUE, border = "red", probability = TRUE, col ="red")
```



Pour chaque jeu de données parmi les 5, on peut regarder l'évolution de la moyenne et l'écart type lors du processus itératif entre lenght et breatdh

```
plot(dm, c("type", "length"))
```



Iteration

```
logit_model <- glm(type ~ breadth + height + face_breadth + face_height, data = d1, family = binomial(li
summary(logit_model)
```

```
##
## Call:
## glm(formula = type ~ breadth + height + face_breadth + face_height,
##     family = binomial(link = "logit"), data = d1)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -49.7508    28.8896  -1.722   0.0851 .
## breadth       -0.2267     0.1235  -1.836   0.0663 .
## height         0.1048     0.1017   1.030   0.3031
## face_breadth   0.2387     0.1528   1.563   0.1181
## face_height    0.4888     0.2426   2.015   0.0439 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 44.361  on 31  degrees of freedom
## Residual deviance: 22.272  on 27  degrees of freedom
## AIC: 32.272
##
## Number of Fisher Scoring iterations: 7
```

- breadth : -0.2267. Chaque augmentation d'une unité de la variable 'breadth' est associée à une

diminution de l'odds ratio de 0.2267, bien que cela ne soit pas statistiquement significatif à un niveau de confiance de 95% (p-value = 0.0663).

- height : 0.1048. Chaque augmentation d'une unité de la variable 'height' est associée à une augmentation de l'odds ratio de 0.1048, mais cela n'est pas statistiquement significatif (p-value = 0.3031).
- face_breadth : 0.2387. Chaque augmentation d'une unité de la variable 'face_breadth' est associée à une augmentation de l'odds ratio de 0.2387, bien que cela ne soit pas statistiquement significatif (p-value = 0.1181).
- face_height : 0.4888. Chaque augmentation d'une unité de la variable 'face_height' est associée à une augmentation de l'odds ratio de 0.4888, et cela est statistiquement significatif à un niveau de confiance de 95% (p-value = 0.0439). Cela suggère que 'face_height' a une influence significative sur la probabilité de 'type'

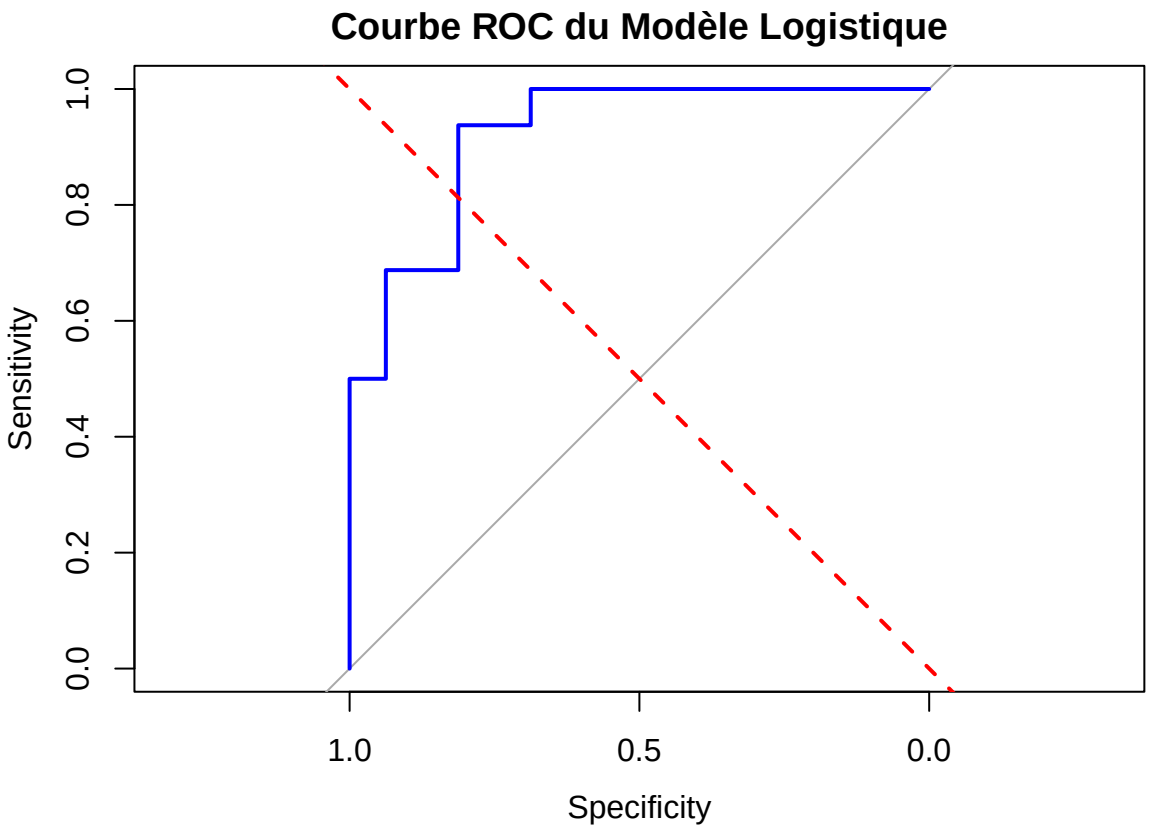
Courbe ROC

```
predictions <- predict(logit_model, d1, type = "response")
roc_obj <- roc(d1$type, predictions)
```

```
## Setting levels: control = A, case = B
```

```
## Setting direction: controls < cases
```

```
plot(roc_obj, main = "Courbe ROC du Modèle Logistique", col = "blue", lwd = 2)
abline(a = 0, b = 1, col = "red", lty = 2, lwd = 2)  # Ligne de référence
```



Calcul du taux de mal-classés avec un seuil de décision de 0.5

```r
threshold <- 0.5
predicted_class <- ifelse(predictions >= threshold, "B", "A")
misclassification_rate <- mean(predicted_class != d1$type)
cat("Taux de mal-classés (seuil de décision = 0.5):", misclassification_rate, "\n")
```

```
## Taux de mal-classés (seuil de décision = 0.5): 0.15625
```